

## Pilot study on deploying a wireless sensor-based virtual-key access and lock system for home and industrial frontiers

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### ABSTRACT

The rise in data processing activities vis-à-vis the consequent rise in adoption and adaptation of information and communication tech related approaches to resolve societal challenges has become both critical and imperative. Virtualization have become the order of the day to bridge various lapses of human mundane tasks and endeavors. Its positive impacts on society cannot be underestimated. This study advances a virtual wireless sensor-based key-card access system with cost-effective solution to manage access to restricted areas within a facility. We seek to integrate virtual key card access, web-access control, solenoid lock integration, and ESP32-controller to create a dependable internet of things (IoT)-enabled access control system. Results show system benefit includes improved security, improved convenience, privacy, efficiency with real-time control capabilities that will allows building administrators to track and manage access to the facility remotely.

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## 1. INTRODUCTION

Today, doors are designed to keep intruders out of both public and private infrastructures with locks neatly installed as security measures to help achieve this [1], [2]. Doors are installed and used in both hotel, homes, and office – to mention a few, as its mundane function is to either grant access to authorized persons, or keep out intruders cum strangers from restricted areas. Thus, facility managers utilize doors as security measure that prevents unauthorized access to unauthorized users [3], [4]. A predominant challenge facing many individuals and businesses today – is the security of both infrastructure, properties and lives [5], [6] as secure platforms require secure protocols as necessary measures to yield high-end user trust and requisite protection canvassed for [7]. Advances in informatics technology has made it imperative to adopt and adapt such emergent techs in almost every facet of life's endeavor; whilst, ensuring a consequent adaptation of the requisite imperative security measures [8]–[10] therein such facilities.

Administrators of such facilities and infrastructure today – have commenced the exploration and adoption of new technology targeted at advancing security protocols and measures that both dissuades

unauthorized access as well as ensure that legitimate users are not prevented access to such workspaces. This has become a crucial design component in modern homes over time [11], [12]. Advances made generally to provision better living from inception, is targeted at utilization of imaginary locks that protects user privacy as well as user's personal property [13], [14]. Its consequent improvement has continued to ensure greater protection overtime. So many of such door security protocols currently in place have also successfully, proven to be somewhat insecure, unreliable and are easily bypassed by adversaries.

With many doors often left unlocked from forgetfulness and other reasons – it is become imperative and critical to provide smart-locks [15], [16] which seeks to explore and exploit the use of embedded systems with internet-capabilities [17] via the use of key-codes, smart-phones, and key-cards – as means to ensure safety and ease user-trust of lock systems, and this agrees with [18], [19]. Also, a majority of devices used in such lock-system [20] are riddled with flaws that adversaries and intruders, also exploit to gain unauthorized access to barred locations [21]. In our effort to close these security gaps, our study employs the use of cryptographic [22] to provide extra layers and improved security for the virtual key-card lock system.

## 2. LITERATURE REVIEW

### 2.1. Review of related literature

Kong *et al.* [23] designed an radio frequency identification (RFID)-based automatic access control system that employed its universal serial bus (USB) as an effective means to communicate/interface with a host computer machine using the peripheral interface controller (PIC) 16f877A [24]. Its graphic user interface program provides functionalities of the overall system such as display of live ID tag transactions, registering ID, deleting ID, recording attendance, and other functionalities. It was deployed using Visual Basic 2010 with feature for registering and deleting ID makes the system more flexible but the system lacks facilities for true user identification such as a camera, and fingerprint scanner [25]. An improvement that can be made to this system is the use of an RFID fingerprint scanner instead of a tag to rule out the possibility of unauthorized access [26], [27]. Yuan and Xu [28] presented an Android-based control system to maintain the security of the home's main entrance and also the car door lock. The system can also control the overall appliances in a room. The mobile-to-security system or home automation system interface is established through Bluetooth. The hardware part is designed with the PIC microcontroller.

Joshi *et al.* [29] extended Joshi and Vaghela [30] for smart home technology using bluetooth in a mobile device. This Bluetooth-based Android smartphones was prototyped, and the hardware design for its door-lock system is the combination of an android smartphone features such as the taskmaster, Bluetooth module as command agent, Arduino microcontroller was used as its controller and data processing center, and solenoid as door lock output. Nasir *et al.* [31] also presented and analyzed the design and implementation of a microcontroller-based home security system using the global system for mobile communications (GSM). It used 3-microcontrollers to extend/expand the functionalities in other peripherals such as light emitting diode (LED), LCD, Buzzer, and a GSM module are responsible for the reliable operation of the proposed security system [31], [32]. Sun *et al.* [3] extended Nasir *et al.* [31] developed two remote monitoring systems using a cell phone with a focus on wider utilization. The first system is designed with an ARM LPC 2148 microcontroller based on commands received from the user's cell phone and presents sensor conditions to the LPC 2148 microcontroller system to send signals via its ports to switch appliances on/off; While, its second system incorporates some additional features like capturing and store of an intruder's images unknown to him/her [33]. Zawislak *et al.* [34] investigated the automatic password-based door lock using electronic technology to build an integrated, fully customized home security system at a reasonable cost. The project is useful in keeping thieves and other sorts of dangers at bay [34], [35].

Kim *et al.* [24] extended Bhavani and Mangla [36] via RFID-based automatic access control system that used a USB as an effective means to communicate/interface with a host computer machine using the PIC 16f877A. They extended its user interface to yield greater functionalities of the overall system such as display of live ID tag transactions, registering ID, deleting ID, recording attendance, and other functions [37]. The embedded system features registration and deletion of IDs makes the system more flexible but the system lacks facilities for true user identification via Computer Vision model view to rule out the possibility of unauthorized access [38], [39]. Designed a password-protected home automation system with an automatic door lock using the Arduino board as controlled by ATmega-328. First, the user combination will be compared with the pre-decided passwords stored in the system memory [40]. If the user's combination matches the password, the door, light, and fan will be unlocked [41]. The system was built to be locked by just pressing a key. It used the Arduino board to interface various peripherals. If the password is matched with a pre-decided password, then the Arduino simply operates the relay to open the lights and fan. The Arduino simultaneously operates a DC motor via a motor driver for operating the door [42]–[44].

## 2.2. Wireless sensor networks

Wireless sensor networks (WSN) are fastidious in their adoption and extension of virtual technology. Virtual techs today, are used by many systems globally to include a variety of automation ranging from virtual remote-based light controllers, smart interfaces and many more. At its core is the embedded system that can adjusted with little modification to actualize virtual assistive techs. These can also be retrofitted to fully utilize our mobile devices capabilities. In all, they raise the quality of life for individuals who use these systems [45], [46]. Due to fast-paced advances in technology, smart key-cards are providing users with a variety of options to creating cost-effective, robust, flexible and low-maintenance virtualization solutions, there has since become a rise in the trend in the adoption and adaptation of such virtualization solution due to their dynamism and high-evolution [47]–[49].

Businesses and homes today, that integrate the use of physical servers with onsite/off-site locations will often profit from low-cost implementation, reduced maintenance cost, improved administration and over-simplification that accompanies a virtualized server databanks and environment [50], [51]. Through such shared resources, virtualization enables the expansion of hardware [52]–[54]. A plethora of restrictions often accompany such virtual systems – one critical component being the dearth of possibilities that can be brought together in one location [55], [56]. Furthermore, there is also the lack of high-security choices. To resolve such problem, we wish to combine all the current (security features, safety features, and monitoring functions) into a single, virtual smart-lock. This will thus, yield a highly-secured system that seeks to bridge the gaps in frontier door security options without conflict – to make our homes safer than before [57]–[59]. Thus, intelligent security systems evolve and are deployed to forestall illicit invasions of user privacy. Study aims to provide security protocols for adoption in a door lock system with a single-key for one-lock phenomenon [60], [61].

## 2.3. Study motivation

The study is motivated as listed below, and seeks to achieve the following:

- a) Security: to ensure that virtual keycard door lock system(s) are secured from unauthorized access, adversary hacking, and tampering – security has become a critical component and challenge facing the development and deployment of the smart virtual keycard door lock system. While, it can be applied to other aspects of our daily endeavour, the secured smart keycard door lock system must aim at provisioning secure protection for user sensitive and personal user information as well as prevent unauthorized access to such secure buildings and facilities [62], [63].
- b) Privacy integration with internet of things (IoTs): a key challenge in the deployment of IoT-based and enabled systems is that of privacy from adversarial attacks, threat cum unauthorized-to-compromised access. To enhance user-trust and privacy, such IoT-based lock mechanisms are linked and connected to the internet via smart mobile devices. This is aimed at ensuring the generated system is robust, productive and innovative [64], [65].
- c) Low-cost energy solutions: deploying for use, virtual keycard lock often makes it energy-efficient, it reduces its implementation cost, and ensures it is at a cheaper cost of maintenance. While, cheaper maintenance is not a panacea for improved system reliability – its use is very restrictive so that only a few clients, individuals or organizations can afford it. Biometric systems often have been found to violate users' privacy as some users often consider them to be personally invasive due to loss of anonymity [66].
- d) A key challenge that is faced in this project is the security and privacy of the IoT systems. Therefore, the paper will present an extensive investigation of the security and privacy of IoT systems seeking to enhance the lock mechanism by connecting it to the internet, making it more robust, productive, and innovative.

The study achieves improved security, user data privacy, improved user-trust, energy efficiency and low-power computation via the deployment of a smart, virtual keycard door lock using the IoT-enabled device(s).

## 2.4. The experimental virtual key-card wireless sensor model

The existing output design for the smart virtual door access primarily revolves around a seamless and efficient user experience. A user successfully taps their near field communication (NFC)-enabled device (smartphone or smartcard) on the NFC reader located near a door, which triggers the system door either lock to unlock as seen in Figure 1. Upon successful access, the door lock emits an audible signal to indicate an unlock state, with immediate feedback to the user. This design ensures the entire access is swift and easy to allow users access their rooms effortlessly [67], [68]. The system generates a backend server access log with these feats: (a) timestamp of access, (b) the used NFC device unique identifier, and (c) the room number accessed. These logs are accessible via the administrative interface, allowing hotel staff to monitor and review access activities in real-time. The output design of the access logs facilitates data analysis, enabling hotel management to gain valuable insights into guest behaviours and occupancy patterns.

The experimental scheme seeks to enhance hotel room access control via the integration of advanced secure, access tokens technology with user-friendly features. With the existing system, the proposed system addresses many of its identified weaknesses with the advent of new functionalities. All of which seeks to improve user experiences and operational efficiency. With a focus on device compatibility, the system allows guests to access their rooms using a wide range of NFC-enabled devices, including smartphones, smartwatches, and smartcards. It uses an advanced two-factor authentication and data encryption as security measures to ensure the utmost protection of user data alongside access to user credentials. Also, it provides a fail-safe that seamlessly integrates these features on to the hotel infrastructure; and thus, guarantees the continuous access control even in the event of system failures or connectivity issues.

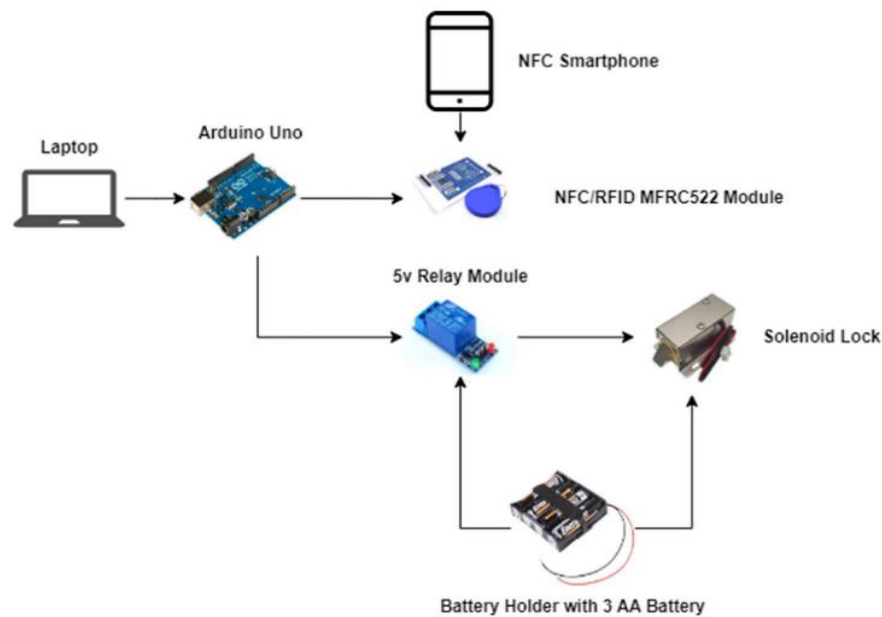


Figure 1. Schematics of the virtual wireless sensor key-card model

The proposed system's intuitive app as in Figure 2, and its administrative interface streamlines the check-in process, and empowers guests to manage their access effortlessly and enabling hotel staff to efficiently handle access permissions and monitor real-time access logs. The proposed virtual key card system promises an enhanced guest experience, heightened security, and improved operational efficiency for modern, tech-savvy hotels as in Figure 2. Which shows circuitry diagram of the proposed system with its structural workings and the schematic diagram of the proposed virtual key-card system as in Figure 3.

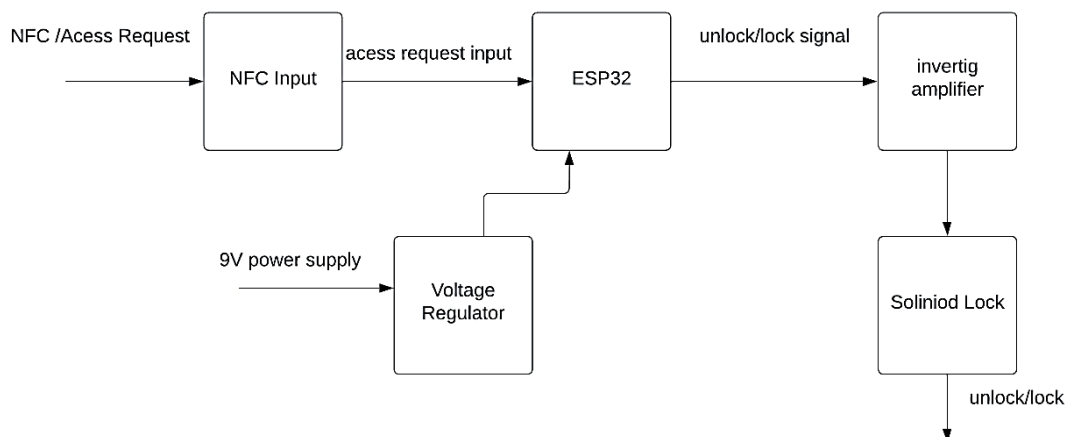


Figure 2. Circuitry diagram of the experimental system

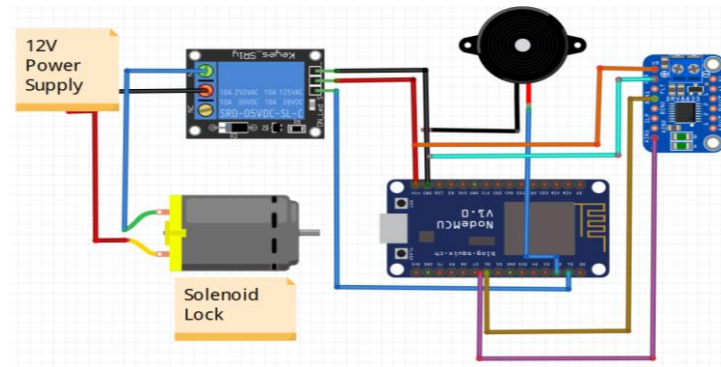


Figure 3. Scematic diagram of the experimental system

Major benefits the proposed system offers will include: (a) an enhanced device compatibility, supports a wide range of NFC-enabled devices for seamless access, (b) advanced security measures, including two-factor authentication and data encryption, ensure robust protection against unauthorized access and data breaches, (c) a fail-safe mechanism that provides continuous access control, even with system failure or network connectivity issues, (d) an equally streamlined integration with existing hotel infrastructure facilitates efficient data synchronization and operations, (e) an intuitive user interface simplifies guest check-ins and eases management of access permissions for both hotel staff and ancillary personnel, (f) improved guest experience with convenient room access using preferred NFC devices, and (g) improved security will improve user-trust, confidence level and safeguards personal information as in Figure 3 which shows the circuit construction of the proposed system architecture.

The proposed system specifications accounted for flexibility, security, robustness and privacy. We used the ESP32 microcontroller due to its powerful capabilities with built-in WiFi and Bluetooth connectivity – both of which are essential for IoT applications. We also integrated a GSM module for usage in areas without internet access. Also, incorporated was the solenoid lock, which will be activated by the microcontroller in response to the appropriate input signals. The system integrated a web-based user-friendly interface that allows users to register and manage their virtual key cards. The system also has robust security features to prevent unauthorized access, including encrypted communication between the microcontroller and the server. The system used a cloud-based server to allow remote management. It uses a card reader for the RFID or NFC tags on the virtual key cards. It yields a display on the web app to provide feedback to users, indicating whether access has been granted or denied. The solenoid lock is designed to withstand tampering, and the entire system is housed in a secure enclosure to prevent unauthorized access.

### 3. FINDINGS AND RESULT DISCUSSION

#### 3.1. Experimental system performance and evaluation

Table 1 shows the performance test result and indices from the system as generated. Table 1 shows performance result of integrating the virtual key-card with IoT and embedded system as used to control a door lock. Results showed system's efficiency and effectiveness as compared to [69], [70]. We also evaluated several key metrics to include access control speed, reliability, and user convenience respectively – as in agreement with [71]. The access control speed was evaluated by measuring the time it took for the system to either grant or deny access to the door lock. The proposed system was found to be fast and responsive, with both granting of and the denial of access to unlock features in less than a second. This quick response time allowed for smooth and efficient access management. This is in agreement with [72]–[74].

Table 1. Performance evaluation for the experimental system

| Test description                                   | Actual results                                                                                              | Remark |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|
| Successful access using authorized NFC tags        | The door unlocks upon presenting an authorized NFC tag                                                      | Pass   |
| Failed access via unauthorized device access       | Door remained lock upon presenting an unauthorized device                                                   | Pass   |
| Failed access using invalid NFC tag                | Door remains locked upon presenting invalid NFC tags                                                        | Pass   |
| Error handling for invalid NFC tag                 | Displays correct error message on the web application and serial monitor upon presenting an invalid NFC tag | Pass   |
| Compatibility with the different types of NFC tags | System is able to read and grant access to different types of NFC tags                                      | Pass   |

Furthermore, we tested for reliability by testing the system's ability to accurately read the virtual key cards and control the locks based on the access rights defined in the web application. The system was found to be reliable, with no instances of incorrect access granted or denied. The locks were also found to be secure and reliable, able to withstand physical stress and external tampering. User convenience was evaluated by assessing the ease of use of the virtual key cards and the web application. The virtual key cards were found to be easy to use, with no special skills or knowledge required. The web app is user-friendly and intuitive, with a simple and straightforward interface, which agrees with [75]–[77].

### 3.2. Discussion of findings

Table 2 shows findings using a variety of the indices from the system. We evaluated using these features to include the following: (a) access control [78], (b) scalability [79], (c) reliability [80], (d) error handling [49], (e) users' usability convenience and satisfaction [81], (f) compatibility with several NFC devices [82], and (g) fault tolerance [83].

Table 2. System test results

| Test metrics                   | Description                                                                                                                                                                          | Result                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access control speed           | The time it takes for the system to grant or deny access to the door to any user                                                                                                     | It takes an average of 1.8 seconds for access to be granted if the tag is used, and 1.1 secs if the web-request is used                               |
| Physical security              | The system's ability to withstand physical stress and external tampering                                                                                                             | Resistant to external tampering and physical stress, with no reported incidents of unauthorized access or damage                                      |
| Reliability                    | System's ability to accurately read the virtual key-card and control the locks based on the access rights defined in the web-application                                             | System had 99.5% accuracy in reading key-card, controlling the locks to lock/unlock states based on access rights defined in the web app              |
| User convenience               | The ease with which a user uses the virtual key-card and web application                                                                                                             | Users found the virtual key cards and web apps quite easy to use and convenient, with no major use issues or complaints                               |
| System fault tolerance         | The system's ability to operate reliably over a prolonged period of time                                                                                                             | The system operated reliably over a prolonged period of time, with no reported incidences of downtime and failure to access the network               |
| System compatibility with NFCs | It describes how compatible system is with the plethora of other devices with NFC capabilities such as laptops, smartphones                                                          | System is compatible with devices and techs that incorporated WIFI, Bluetooth and NFCs                                                                |
| Error handling                 | System's ability to support a large number of users and access points, handle concurrent requests, invalid NFC tags and WiFi connectivity issues                                     | The system supports a large number of client and user access points with no performance degradation and security issues reported                      |
| Scalability                    | Ability to support large number of users and access points, handle concurrent requests, invalid NFC tags and WiFi connectivity issues and not compromise performance and/or security | System was found to support large number of client and user-access points without any form of performance degradation and/or security issues reported |

The system yields a 99.5% accuracy in reading the virtual key-card and in controlling the lock/unlock states based on defined access-rights in the web app. The system yields fault-tolerance and is compatible with many NFC tags and accompanying devices, to agree with [84], [85]. We successfully used an ESP32, a solenoid lock, a web API, and a web app to manage the access of authorized personnel [86]–[88]. It yields several benefits like increased security, efficiency, and convenience, which agrees with [89]. It also eliminates the challenges associated with traditional key like risk of lost or stolen keys, the inconvenience of having to carry physical keys, and the lack of real-time access control and monitoring. It allows authorized personnel to access designated areas without the need for physical keys, reducing the risk of lost or stolen keys [90], [91].

The system provides real-time access control and monitoring, allowing administrators to track and manage access to the facility remotely [92]–[94]. This study can be advanced to include additional features, such as facial recognition and voice recognition, to enhance the security of the system. The study has also contributed by demonstrating the inherent potentials in the use of IoTs/embedded systems to provide innovative solutions to complex problems in various industries.

## 4. CONCLUSION

The system yields a cost-effective solution to manage user access integrating IoTs to create a comprehensive access control system. Its many benefits over traditional key includes better security, user data privacy, system efficiency, and user convenience. The system also provides real-time monitor and control capabilities that will allow administrators to track and manage access to the facility remotely. And in turn, enhancing system's security and efficiency.

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